A Formal Study of the Doppler Tolerance of Costas and Sudoku Waveforms

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***Abstract*—We apply a new, formal measure of Doppler tolerance to frequency-hopping waveforms made from Costas arrays and Sudoku-based permutations. We first give a combinatorial structure theorem for Sudoku permutations that facilitates waveform analysis. We form statistical profiles of Doppler tolerance envelopes computed from thresholding narrowband ambiguity functions of these waveforms and compare against those from randomly-chosen permutations. As functions of temporal offsets, the statistical profiles indicate a wide range of Doppler tolerance, suggesting that permutation adaptivity is a worthwhile degree of freedom to exploit for moving target indication in frequency-challenged environments. We find that Sudoku-based waveforms behave much like randomly-chosen permutations with respect to Doppler tolerance.**

I. Introduction

The Doppler tolerance of the transmit waveform of a radar system is a key performance consideration for radar scientists and engineers. A target with a significant Doppler shift (a consequence of radial speed, wavelength, as well as pulse repetition frequency) can become impossible to detect. The radar system can have an undesirably low matched filter response if the transmit waveform is not tolerant to a Doppler shift.

Doppler tolerance has had various intuitive but imprecise or precise yet incomplete formulations in the literature. Among the latter, some do not take filter mismatch into account. One widely-held belief has been that linear frequency modulated (LFM) chirp waveforms are Doppler tolerant.

Setlur et al. [1] surveyed notions of Doppler tolerance and develop a formal, general expression for Doppler tolerance using thresholded ambiguity functions as a maximal frequency offset. Their narrowband metric is defined in terms of a connectedness parameter ϵ_0 for regions of the hard-thresholded narrowband ambiguity function, the time delay τ at which the Doppler tolerance is to be computed, and the threshold α of the narrowband ambiguity function (the parameter α depends directly on signal-to-noise ratio.) For small time-bandwidth products, they showed that the LFM waveform is only as Doppler tolerant as an unmodulated rectangular pulse.

In this paper, we start a comparison of the formal Doppler tolerance metric for frequency hopping waveforms of the new Sudoku waveforms of Narayanan, Bufler, and Leshchinskiy [2], Costas arrays, and randomly-chosen permutations. We first give a combinatorial characterization of Sudoku permutations that streamlines such investigations. At a given temporal offset, we see that there is substantial variance in Doppler tolerance across permutations, indicating that transmit pattern adaptivity for Doppler tolerance is a worthwhile degree of freedom. In general, we find that the Doppler tolerance of Sudoku waveforms behaves much like that of randomly-chosen permutations. Since Sudokus are plentiful, this indicates that switching within a large library of them can be used to increase low-probability-of-intercept/low-probability-of-detection (LPI/LPD) properties of a radar system.

In our companion paper [3], we consider pulse trains of Sudoku waveforms, the effects on Doppler tolerance of pulse repetition interval (PRI) staggering to control the bed-of-nails in the narrowband ambiguity function of a pulse train, as well as Sudoku waveforms arising from higher-order Sudoku (to accommodate a variety of time-bandwidth products) as well as some Sudoku variants.

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II. Signal Model and Narrowband Ambiguity Function

Following [4, Section 2], we present a mathematical model for the complex envelope of a PRI-staggered pulse train of a frequency-hopped signal and an expression for its narrowband ambiguity function specialized herein and used in our companion paper [3].

Define a pulse train of N pulses as

$$s(t) = \sum_{n=0}^{N-1} s_n(t - nT_r - p_nT), \tag{1}$$

where $s_n(t)$ is the complex envelope of the n-th transmitted pulse, T_r is the nominal PRI, and T is the pulse-position-modulation offset duration quantization. In (1), p_n adds an offset relative to the n-th pulse position to allow for pulse staggering. We assume that $s_n(t) = 0 \; \forall \; t \notin [0, T]$. Define

$$P = \max_{0 \le n \le N-1} p_n. \tag{2}$$

We require that $T_r \geq 2PT$ so that only one pulse is transmitted in each PRI and cross-ambiguity function contributions to the pulse train do not overlap.

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In this paper, we set $N = 1$ to carry out single-pulse analyses. In our companion paper [3], we use $N > 1$ to consider staggered pulse trains. In both papers, we are interested in frequency-hopping signals (a form of frequency-jump-burst) which we define up to a real constant. Let

$$s_n(t) = \exp\left(j\left(\phi_k + 2\pi\left(f_0 + \frac{f(k)}{h}\Delta f\right)t\right)\right), \quad (3)$$

where j denotes $\sqrt{-1}$ as usual, $\phi_k \in [0, 2\pi)$ are phases chosen to analytically enforce phase continuity, f_0 is the carrier frequency, $f : [h] \to [h]$ is a frequency hopping mapping with h equal to the number of frequency hops and $[h] = \{1, 2, \ldots, h\}$, B denotes pulse bandwidth (h hops), and $t \in \left[\frac{k-1}{h}T_p, \frac{k}{h}T_p\right]$ is fast-time step interval with $T_p = h\tau_1$ equal to pulse-length, τ_1 equal to tone duration. The mapping f is taken to be a permutation in this paper.

We acknowledge that frequency-hopping is typically done in pulse trains ($N > 1$). Indeed, pulsed-Doppler systems use pulse trains thousands of pulses long to sharpen Doppler resolution. In this paper, we begin the study of the Doppler tolerance of Sudoku waveforms by presenting a structural result and giving single-pulse analyses.

Write the narrowband auto ambiguity function of the narrowband signal $s(t)$ as

$$\chi(\tau, \nu) = \int_{-\infty}^{+\infty} s(t)s^*(t-\tau)e^{j2\pi\nu t}dt \quad (4)$$

and the cross ambiguity function of the narrowband functions $s_n(t)$ and $s_m(t)$ as

$$\chi_{n,m}(\tau, \nu) = \int_{-\infty}^{+\infty} s_n(t)s_m^*(t-\tau)e^{j2\pi\nu t}dt. \quad (5)$$

Substituting (1) into (4) and grouping terms gives

$$\begin{aligned}\chi(\tau, \nu) = &\sum_{n=0}^{N-1} e^{j2\pi\nu(nT_r+p_nT)}\chi_{n,n}(\tau, \nu) \\ &+ \sum_{k=1}^{N-1}\sum_{n=0}^{N-1-k} e^{j2\pi\nu(nT_r+p_nT)}\chi_{n,n+k}(\tau_2, \nu) \\ &+ \sum_{k=1}^{N-1}\sum_{n=0}^{N-1-k} e^{j2\pi\nu[(n+k)T_r+p_{n+k}T]}\chi_{n+k,n}(\tau_3, \nu),\end{aligned} \quad (6)$$

where

$$\tau_2 = \tau + kT_r + (p_{n+k} - p_n)T \quad (7)$$

$$\tau_3 = \tau - kT_r - (p_{n+k} - p_n)T. \quad (8)$$

The first summation in (6) shows the weightings of the auto ambiguity functions of the component pulses of the pulse train. The second and third summations are due to the cross-ambiguities of component pulses. Equation (6) shows that pulse staggering affects both the weighting of main lobes along the Doppler (ν) axis and creates shifts of the cross ambiguity functions along the delay (τ) axis. We refer the reader to [5] for further analysis and discussion of narrowband ambiguity functions of PRI-agile signals.

III. Sudokus and Sudoku Waveforms

A Sudoku is an order-9 Latin square with the additional constraint that each of the nine 3×3 canonical squares contains the numbers 1 through 9 [6]. One obtains a Sudoku permutation matrix from a Sudoku by zeroing out all integers besides a fixed number and recording the positions of the non-zero entries [2, Figure 1]. Narayanan, Bufler, and Leshchinskiy [2] propose using Sudoku permutation matrices to generate frequency-hopping transmit patterns and explore applications to antennas with coauthor Sherbondy [7]. They demonstrate [2] that the resulting single-pulse Sudoku waveforms have low average peak sidelobe ratio (PSLR), have fine range and Doppler resolution, and low cross-ambiguity, indicating that signals derived from fixing different digits do not interfere with one another when operating in close proximity and are also beneficial for use in multiple-input-multiple-output (MIMO) radar applications. Figure 1 illustrates the narrowband ambiguity function and auto-coincidence surface for the Sudoku permutation 1 4 7 2 6 8 5 3 9.

Sudoku properties allow for switching the Sudoku frequency hopping transmit pattern to enjoy noise-like behavior. The Sudoku constraint results in Sudokus having higher Shannon entropy (a measure of disorder for a matrix) than randomly-chosen matrices [8]. It is known [9] [10] that there are exactly $6,670,903,752,021,072,936,960$ Sudokus and that, up to symmetry, there are $5,472,730,538$ [11] Sudokus. An $n^2 \times n^2$ Sudoku has the analogous property for each of the n^2 $n \times n$ canonical squares and must exist (Theorem III.1) for all positive integers n. A rich variety of puzzle variants and sizes now exists [6].

The sidelobe structure of the (single-pulse) narrowband ambiguity function χ of a Sudoku waveform varies considerably with Sudoku (Tables II,I). On the other hand, all Costas arrays have a thumbtack narrowband ambiguity function. All known Costas arrays from order 3 to 1030 are available online [12] and Costas arrays of infinitely many orders exist. Selection of order for target detection and then Costas array is appropriate. Setting order to be the nearest integer to the square root of the time-bandwidth product is reasonable (see [13, Section 3] for more on these topics). Sudoku waveforms work well with a time-bandwidth product around $9^2 = 81$ and larger Sudokus can be used for greater time-bandwidth products.

A. The structure of Sudoku permutations

The following structural result facilitates analytical studies by removing the need for backtracking computations to create tables of Sudoku permutations.

Theorem III.1. *Sudoku permutation matrices are precisely those permutation matrices that have the additional property that there is exactly one 1 in each of the 3×3 canonical squares.*

Proof: Clearly, a Sudoku permutation matrix has the property that there is exactly one 1 in each of the 3×3 canonical squares.

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Suppose we have a permutation matrix such that there is exactly one 1 in each of the 3×3 canonical squares. We present how to fill in each of the 3×3 subsquares to create a Sudoku, so the given permutation matrix is the Sudoku permutation matrix attained by zeroing out all integers except the 1s. Essentially, we think of each 3×3 subsquare as a gluing diagram for a torus, placing a 2 directly "below" the 1 and a 3 directly "below" the 2, where "below" means in the same column in the first row if the number is to be placed "below" the third row. Next, place a 4 directly "to the right" of the 1, a 5 "below" the 4 and a 6 "below" the 5, where "to the right" means in the same row in the first column if the number is to be placed "to the right" of the third column. Finally, place a 7 "to the right" of the 4, an 8 "below" the 7 and a 9 "below" the 8. The set of numbers $\{1,2,3\}$ (and $\{4,5,6\}$ and $\{7,8,9\}$) will be in the same column of any 3×3 subsquare and will be in unique columns since the 1s (and the 4s and the 7s) are in unique columns. The set of numbers $\{1,4,7\}$ (and $\{2,5,8\}$ and $\{3,6,9\}$) will be in the same row of any 3×3 subsquare and will be in unique rows since the 1s (and the 2s and the 3s) are in unique rows. Thus, the resulting matrix is a Sudoku, completing the proof. ■

Note that a permutation matrix of order 9 has the additional property that there is exactly one 1 in each of the 3×3 canonical squares precisely if the corresponding permutation has exactly one number from $\{1,2,3\}$, one number from $\{4,5,6\}$ and one number from $\{7,8,9\}$ among its first 3 numbers (and its second 3 numbers and its third 3 numbers). We build such a permutation by first ordering each of the 3 sets of numbers in $(3!)^3$ ways and then ordering the sets of first, second and third numbers from these orderings in $(3!)^3$ ways. Thus, Theorem III.1 implies that there are $(3!)^6 = 46,656$ Sudoku permutations. We pre-computed all of them for waveform analysis studies presented herein. Theorem III.1 generalizes to give $(n!)^{2n}$ length-n^2 Sudoku permutations.

B. Combinatorial Analysis of Sudoku Ambiguity Functions

We partitioned the $9! = 362,880$ order-9 permutations into 3 sets: 46,650 Sudoku permutations that are not Costas arrays, 760 order-9 Costas arrays, and the remaining permutations which we refer to these as random although we have an exhaustive list of them. We analyzed and compared the sidelobe levels and area around the central peak of the auto-coincidence functions of Sudoku and random order-9 permutations.

1) Auto-Coincidence Analysis: The auto-coincidence matrix or co-hit array is given by performing a 2-D correlation of the frequency-hop permutation matrix with itself. Examining the resulting co-hit array allows each permutation to be evaluated for performance by looking at the highest collision besides the central peak. This is a single-pulse analysis but it contributes to pulse trains via the first sum of (6).

The central peak of a co-hit array will always equal the order of the permutation, in this case 9. Finding the second highest number of collisions in the co-hit array will give the highest number of coincidences within a given permutation. All Sudoku permutations were examined by taking each of the nine permutations corresponding to the digits 1-9 and performing their co-hit array analysis on each digit. This analysis is performed for Sudoku and random permutations to compare their results–see Table I.

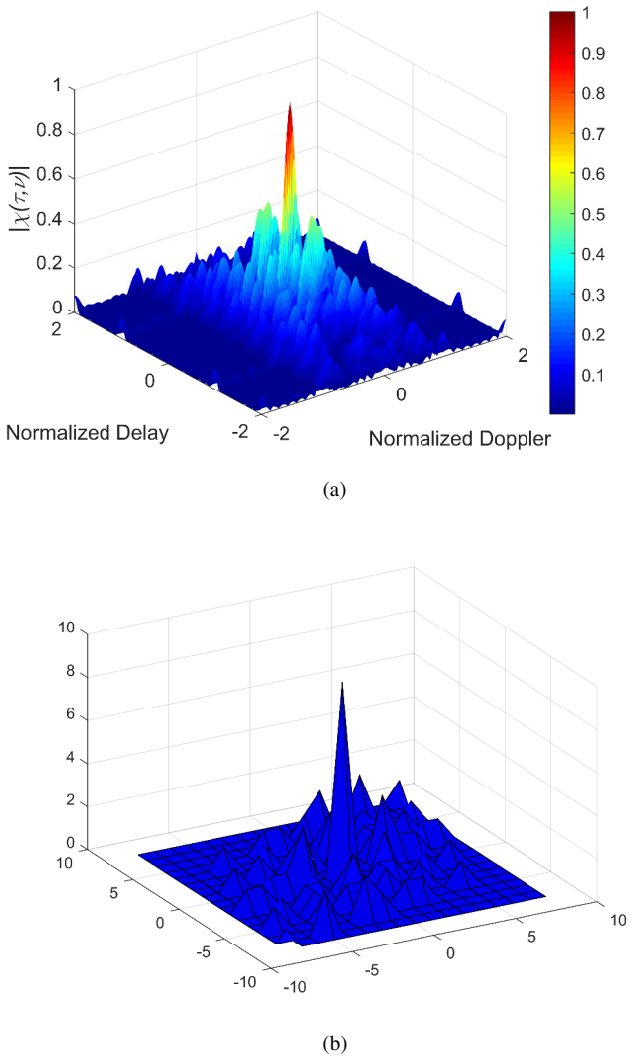


Fig. 1: Single-pulse Ambiguity surfaces for the Sudoku permutation 1 4 7 2 6 8 5 3 9. (a) Narrowband ambiguity function, (b) Auto-coincidence function.

In general, the probability of having a lower number of collisions is higher for Sudoku permutations than just for a random permutation of numbers. Notice that no permutations resulted in a max sidelobe of one due to the fact that all Costas permutations were removed from the analyses. The random permutations even showed coincidences as high as eight. Using a random permutation, there is a possibility of having a linear subsequence which results in a linear frequency modulation having linear decreasing coincidences along the ridge of the ambiguity plane.

2) Pedestal Region: The central pedestal region is of interest as a clearer central region provides better target detection

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TABLE I: Auto-coincidences of Sudoku and random permutations.

Max Sidelobe	2	3	4	5	6	7	8
Sudoku	49.93%	42.41%	6.67%	0.92%	0.069%	0%	0%
Random	47.34%	43.67%	7.95%	0.93%	0.093%	0.0076%	0.006%

TABLE II: Area Around Pedestal.

Central Square	3×3		5×5		7×7	
Max Sidelobe	Sudoku	Random	Sudoku	Random	Sudoku	Rrandom
0	32.0146 %	10.3625 %	0.0043 %	0.0 %	0.0 %	0.0%
1	51.5464 %	43.2438 %	23.4880 %	8.4193 %	1.5636 %	0.7238%
2	14.6220 %	32.9205 %	59.4845 %	57.8502 %	59.9055%	54.5323%
3	1.7354 %	10.6845 %	14.8024 %	27.3127 %	32.2766%	36.6219%
4	0.0816 %	2.3393 %	1.9974 %	5.5153 %	5.3351 %	7.1181%
5	0.0 %	0.3904 %	0.1976 %	0.8037 %	0.8505 %	0.9025%
6	0.0 %	0.0532 %	0.0258 %	0.0906 %	0.0687 %	0.0932%
7	0.0 %	0.0051 %	0.0 %	0.0076 %	0.0 %	0.0076%
8	0.0 %	0.0006 %	0.0 %	0.0006 %	0.0 %	0.0006%

for closely-spaced targets and fewer sidelobes masking targets. The central peak has a value of order n in the auto-coincidence surface. We computed the auto-coincidence functions of all Sudoku and randomly-chosen permutations and then the max number of coincidences in concentric squares centered on the peak. The squares were 3×3, 5×5, and 7×7 grids with the tabulated results given in Table II.

The tabulated results show that Sudoku puzzles have significantly lower numbers of collisions than random permutations, especially near the pedestal for areas of 3×3 and 5×5. Moving further out and looking at the 7×7 grid, the results are nearly the same. Comparing the Sudoku and random permutations to Costas arrays would show that Costas would have a max of one coincidence for all areas searched, in that case it would be ideal to look at which permutations provide the cleanest pedestal area before a one is encountered.

IV. A Formal Measure of Doppler Tolerance

A formal measure of Doppler tolerance is put forth in [1, Section III.A] in terms of hard-thresholded ambiguity functions. We now relate the narrowband formulation of this concept. Define the hard-thresholded narrowband ambiguity function as

$$\chi_\alpha^{xx}(\tau,\nu) = \begin{cases} \chi(\tau,\nu) & \frac{|\chi(\tau,\nu)|}{\max_{\tau,\nu}|\chi(\tau,\nu)|} \geq \alpha \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

and notice the normalization of χ that it performs. Suppose that $\chi_\alpha^{xx}(\tau,\nu)$ divides the τ-ν plane into M disjoint non-zero regions. Let the mainlobe region be E_0^α and the sidelobe regions be E_j^α, for $j = 1, 2, \ldots, M-1$. Due to symmetry across the origin ($|\chi(\tau,\nu)| = |\chi(-\tau,-\nu)|$ for all $(\tau,\nu) \in \mathbb{R}^2$), the set consists of identical sets shifted in the τ-ν plane. The authors introduce connectedness parameters ϵ_j, $j = 1, 2, \ldots, M-1$, which measure the minimum distance of the thresholded sidelobe regions to the mainlobe region as

$$\epsilon_j = \text{dist}(E_j^\alpha - E_0^\alpha) = \min\left\{ \sqrt{(\tau_j-\tau_0)^2 + (\nu_j-\nu_0)^2} \,\middle|\, (\tau_j,\nu_j) \in E_j^\alpha, (\tau_0,\nu_0) \in E_0^\alpha \right\}. \tag{10}$$

The authors use a disconnectedness parameter ϵ_0 to exclude distant sidelobes from consideration by defining an inclusion set E^α as follows:

$$E^\alpha = E_0^\alpha \bigcup_{j=1}^{M-1} G_j^\alpha \tag{11}$$

$$G_j^\alpha = \begin{cases} E_j^\alpha & \text{if } \epsilon_j \leq \epsilon_0 \\ \varnothing & \text{otherwise.} \end{cases} \tag{12}$$

The Doppler tolerance $\mathcal{D}_{\epsilon_0}^{xx}(\tau,\alpha)$ of a signal $x(t)$ for a specified ϵ_0 is defined as

$$\mathcal{D}_{\epsilon_0}^{xx}(\tau,\alpha) = \begin{cases} |\sup_\nu F_\tau^\alpha - \inf_\nu F_\tau^\alpha| & \text{if } F_\tau^\alpha \neq \varnothing \\ 0 & \text{if } F_\tau^\alpha = \varnothing \end{cases} \tag{13}$$

$$F_\tau^\alpha = \{\nu | (\tau,\nu) \in E^\alpha\}. \tag{14}$$

In this paper we regard ϵ_0 as $+\infty$ in order to not exclude any sidelobes.

V. Computational Findings

In this section, we present computational Doppler tolerance findings in terms of Doppler tolerance $\mathcal{D}_{+\infty}^{ss}(\tau,\alpha)$ and resulting statistics as the frequency hop permutation f is varied. Although parameter choices help determine the findings, we know from Section IV that the general nature of the sidelobes helps determine Doppler tolerance. In this paper we fix representative parameters for an X-band airborne medium-PRF (pulse repetition frequency) moving target indication radar in Section V-A. We present analyses in Section V-B and discuss use of multiple pulses (for ambiguity resolution) and PRI staggering in Section V-C

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TABLE III: Parameter Settings

Parameter	Description	Value
N	Number of pulses	1 pulse
ΔR	Range Resolution	10 m
h	Number of hops	9
T_p	Pulse length	5.4 μs
B	Pulse bandwidth	14.99 MHz
f_0	Carrier frequency	10 GHz
f_S	A/D sampling rate	32 MHz
d	Duty factor	0.10
f_{PRF}	Pulse repetition frequency	18.51 kHz

A. Parameter Settings

Parameter settings are in Table III. Apart from the number of pulses and the frequency-hopping permutation f, the key parameter is desired range resolution ΔR, which we set to 10 m. We enforce Nyquist sampling and enforce power-of-two sampling (for Fast Fourier Transform (FFT)-based implementation) and assume X-band operation at $f_0 = 10$ GHz.

Our analyses do not use any aperture weighting/apodization or consider minimum detectable velocity. Detection preprocessing such as that of [14] is needed for phase fluctuations interfering with coherent integration of target returns.

We use $h = 9$ hops in order to compare Doppler tolerances for Costas arrays, Sudoku permutations, and randomly-chosen permutations. Nyquist order selection tells us that we should use a time-bandwidth product of $h^2 = 81$ (see [13, Section III] for a discussion of order selection strategies). The two-way time delay τ_{2way} corresponding to this resolution is thus $\tau_{2way} = \frac{2\Delta R}{c}$, where c is the speed of light. Thus, the tone duration τ_1 in the narrowband ambiguity function is given by $\tau_1 = \frac{2h\Delta R}{c}$. We set pulse length T_p and bandwidth B to be

$$T_p = h\tau_1 \tag{15}$$

$$B = \frac{h}{\tau_1}. \tag{16}$$

The analog-to-digital sampling rate f_S must be at or above the Nyquist rate to avoid aliasing. We enforce sample counts in processing to be a power of 2 for the sake of FFT-based processing. Thus, we also require that the number of samples per tone time τ_1 be a power of 2. It follows from (16) that the number of samples must be the lowest power of 2 that is at least n. We select the next power of 2 due to complex sampling. Thus, we have

$$f_S = \frac{2^{\lceil \log_2 n \rceil + 1}}{\tau_1}. \tag{17}$$

We set the permutation f from Beard's online database of Costas arrays [12]. There are 760 Costas arrays of order $h = 9$. For comparison we randomly selected 760 Sudoku permutations from the set of Sudoku permutations described in Section III and randomly generated 760 non-Sudoku, non-Costas permutations.

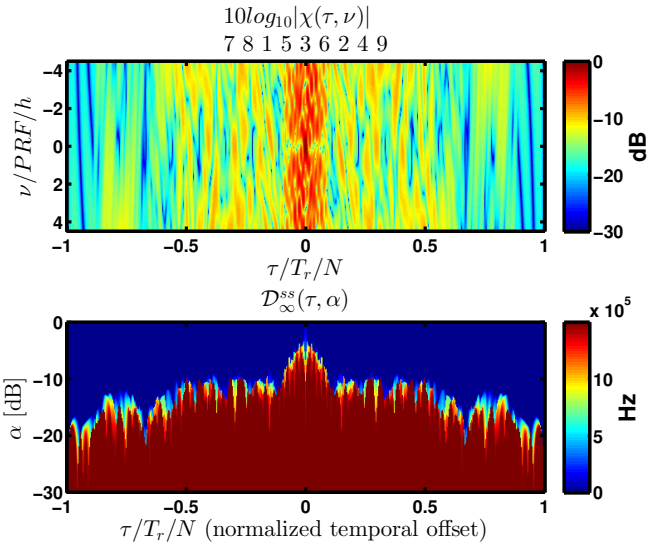


Fig. 2: Narrowband ambiguity function (top) and corresponding Doppler tolerance matrix.

The τ values for which we compute $\chi(\tau,\nu)$ are the τ values at which we can compute $\mathcal{D}^{ss}_{+\infty}$ and the ν values for which we compute $\chi(\tau,\nu)$ help to specify $\mathcal{D}^{ss}_{+\infty}$. We use

$$\tau \in \{-NT_p, -NT_p + \Delta\tau, \ldots, +NT_p\} \tag{18}$$

$$\nu \in \left\{-\frac{h^2 f_{PRF}}{2N}, -\frac{h^2 f_{PRF}}{2N} + \Delta\nu, \ldots, +\frac{h^2 f_{PRF}}{2N}\right\} \tag{19}$$

where $f_{PRF} = dT_p$, $\Delta\tau = \frac{1}{f_S}$ and $\Delta\nu = \frac{c}{2f_0 T_p}$. Hence,

$$0 \leq \mathcal{D}^{ss}_{+\infty}(\tau,\alpha) \leq \max\nu - \min\nu = \frac{h^2 f_{PRF}}{N}. \tag{20}$$

B. Analysis

A representative narrowband ambiguity function together with Doppler tolerance matrix are shown in Figure 2. In order to track hard thresholds of interest, we record minimal and maximal Doppler tolerance envelopes. For each permutation f and each τ, we compute

$$\mathcal{S}_{\alpha}(\tau) = \max\left\{\alpha : \mathcal{D}^{ss}_{+\infty}(\tau,\alpha) = \frac{h^2 f_{PRF}}{N}\right\} \tag{21}$$

$$\mathcal{S}^{\alpha}(\tau) = \min\left\{\alpha : \mathcal{D}^{ss}_{+\infty}(\tau,\alpha) = 0\right\}. \tag{22}$$

Thus, in the bottom plot of Figure 2, $S_{\alpha}(\tau)$ and $S^{\alpha}(\tau)$ indicate the boundaries with blue and dark red colors, respectively, for the temporal offset τ. We compute minimum (m), mean (μ), and maximum (M) statistics across permutations for a fixed temporal offset τ as follows:

$$m = m_{f,\alpha}(\tau) = \min_{f}\left(\mathcal{S}^{\alpha}(\tau)\right) \tag{23}$$

$$\mu = \mu_{f,\alpha}(\tau) = E_f\left[\mathcal{S}^{\alpha}(\tau)\right] \tag{24}$$

$$M = M_{f,\alpha}(\tau) = \max_{f}\left(\mathcal{S}^{\alpha}(\tau)\right) \tag{25}$$

In comparisons for each statistic, larger values of m, μ, and M for fixed τ indicate greater Doppler tolerance. The results are shown in Figure 3. Notice:

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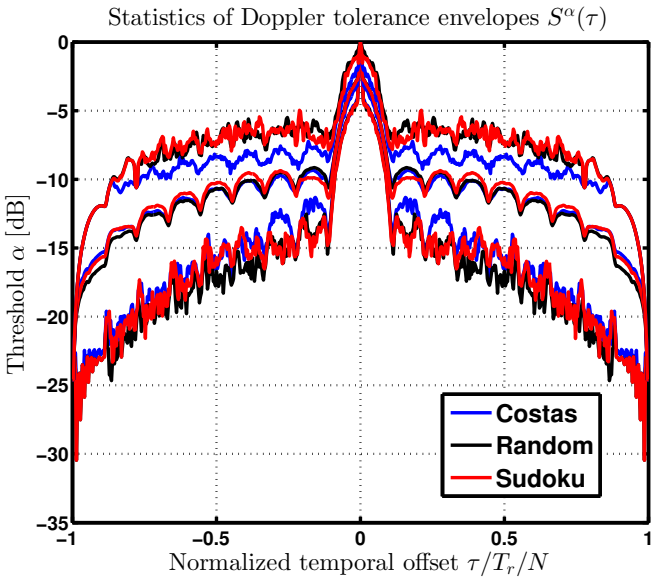


Fig. 3: Statistics of Doppler tolerance envelopes: m (bottom), μ (middle), and M (top) for Costas arrays (blue), random permutations (black), and Sudoku permutations (red).

- The red and black plots of m, μ, and M show that Sudoku permutations closely match the average Doppler tolerance of randomly-chosen permutations.
- The three M plots show that order-9 Costas arrays exist that reach maximal Doppler tolerance at thresholds about 2 dB lower than any Sudoku and random-chosen permutations do for τ such that $|\frac{\tau}{T_r N}| \in [0.1, 0.8]$.
- The three m plots show that order-9 Costas arrays exist that reach maximal Doppler tolerance at thresholds about 2 dB higher than any Sudoku and randomly-chosen permutation do for most τ with $|\frac{\tau}{T_r N}| \in [0.1, 0.3]$.

Statistics computed from $\mathcal{S}_\alpha(\tau)$ are similar to those from $\mathcal{S}^\alpha(\tau)$ and are not discussed.

C. Discussion

It is well-understood (see, for instance, [15]) that the narrowband ambiguity function of a uniform pulse train exhibits a "bed of nails." By definition, the Doppler tolerance metric $\mathcal{D}^{xx}_{+\infty}$ regards these "nails," which are due to ambiguities, as indicative of non-trivial Doppler tolerances over small intervals of τ at high thresholds. The effect of PRI staggering on the narrowband ambiguity function is more elaborate. It is described by [16, pp. 157-158]. Our companion paper explores these topics [3].

VI. Conclusion

In this paper, we analyze and compare the single-pulse Doppler tolerance of three kinds of waveforms for a notional X-band frequency-hopping MTI radar system: Sudoku waveforms [2], Costas arrays, and random permutations. One obtains a Sudoku waveform from a solved Sudoku puzzle by zeroing out all of the digits other than a fixed number. We give a structural result characterizing the resulting permutations that facilitates analysis. We use a new and general Doppler tolerance metric of Setlur et al. [1] that takes into account temporal offsets and thresholding and expresses Doppler tolerance as a maximal frequency offset found from hard-thresholding the ambiguity function. We find that Doppler tolerance across temporal offsets of Sudoku waveforms to be very similar to that from use of random permutations and varies more widely than that from Costas arrays. Our findings support interest in Sudoku waveforms as noise waveform surrogates and indicate the utility of adapting on transmit to boost target detection as well as LPI/LPD performance.

Acknowledgments

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